Samsung plans to use Mediatek's SoC

type of bus is crucial to the implementation of the system and its performance as we shall see in the later chapters.

Why do SoCs exist?

We know what a powerful system the Raspberry Pi is. It is your own DIY mini-computer and it can be programmed to do tasks which were considered huge a mere decade ago. A good example of this is the guy who programmed the entire firmware of the Sony PlayStation One to the RPi – complete with USB controllers and everything. The existence of such a device would surely blow the minds of the developers in the 80s who earned their bread by decreasing memory requirements by the KB. But the driving force behind the Raspberry Pi is an under-appreciated SoC, the Broadcom BCM2835.

Do not be fooled by their tiny size. SoCs support even multi-core processors and GPUs. In fact this field is where the major breakthrough is happening these days. So how is an SoC any different from a motherboard? The answer to this question is quite simple. The motherboard houses different chips for different components. On the other hand, SoCs have all these components on a single chip.

Another special feature of SoCs is that they are super low on power consumption. This is the one feature that makes them irreplaceable parts on devices like smartphones and mini-computers. As a matter of fact, you just need 5V supply to run the RPi via Micro USB. That's effectively a computer powered by a USB phone charger.